

The Icarus costume is not a hop-1 identity switch

A note on the fly_icarus chain. Not a result. Locks stay on the trees.

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Index of locks: <https://gist.github.com/martialsystems/12835f747d6360781f3cc7f91f243178>

Locks stay on the trees. Index: gist 12835f74. Object of study for fly_icarus is constrained toy dynamics on named types. Hop-1 trees from fly_p1_sign onward are connectome measurements. Falsifier on the 11-cell slice: condition 3 not matching condition 1 when HD-on / cVA-off.

Abstract

Question: does a male P1 slice treat a frozen Icarus-from-male as a female? Icarus-from-male keeps male wiring. The object is fat, wingless, and grounded.

On an 11-cell published-sign row, condition 3 matches condition 1 when 7,11-HD is on and cVA is off. Schema w_crit = -1.5262. Template w[P1, DA1] = -1.8. @2cf5fd6.

Hop-1 signed counts onto 88 pC1-coexpress cells: LC10a +606, DA1 +62 ACh, HD -3, ppk23 0. hd_on_cva_off_ns is false. 3 tracks 1 on LC10a. @45aa064.

mAL GABA onto those 88 is -8810. Folded onto the cVA slot, that dump is past w_crit. DA1 hop-1 onto mAL_m* is 0. Numbers @e16856c. Sentence @41437dc.

DA1 reaches 90 of the 93 mAL GABA IDs at hop-2 through type-LH (8773 of 8810). Hop-1 DA1 onto those IDs is 3 of 8810. @de95257.

On the 161 LH cells in that hop, f-bar is 0.099 of all input and 0.703 of uniglomerular PN input. LHAV4c2 carries 1012 of 3301 LH-to-mAL synapses with 5 DA1 synapses. @01155c3.

--unfreeze, --female-brain-icarus, and --n 1000 stay stubbed.

Keywords

Drosophila melanogaster; P1; Icarus; cVA; 7,11-HD; LC10a; mAL; MaleCNS; hop-1; hop-2.

1. Question

Does a male P1 slice treat a frozen Icarus-from-male as a female?

Icarus-from-male keeps male wiring. The object is fat, wingless, grounded.

2. Locks

Schema. 11-cell published-sign row. 3 matches 1 when 7,11-HD is on and cVA is off. Fat, wingless, pinned contact, and female cuticle leave DA1 in place. Schema w_crit = -1.5262 on the 3d pin. Template w[P1, DA1] = -1.8. Song bouts = 1: latch on at step 36, dt = 0.05. @2cf5fd6, logs/p1_contact_s1.json, logs/p1_da1_dose_s1.json.

Hop-1 map. Same battery, MaleCNS hop-1 onto 88 pC1-coexpress cells. LC10a +606. DA1 +62 acetylcholine. HD -3. ppk23 0. ORN_DA1 to DA1 PN is +32,016. Direct ORN to P1 is 0. Every row sings. 3 tracks 1 (d31 = 0.0) because both objects drive LC10a. 3b P1 is 0.9421, above condition 1. hd_on_cva_off_ns is false. Seeds 2 and 3 match. @45aa064, logs/p1_sign_s1.json.

mAL dump. Hop-1 GABA from type-mAL onto those 88 is -8810. Folded into w[P1, DA1] on the parent LC10a scale, the cVA slot is -25.9842, past parent w_crit = -1.539. DA1 hop-1 onto mAL_m* is 0; onto two mALB1 cells it is 96 ACh. hd_on_cva_off_ns is true on that folded row because the slot was given -25.9842. Numbers @e16856c, logs/p1_mal_s1.json. Sentence @41437dc.

Drive. The 93 mAL GABA cells that make the -8810: DA1 hop-1 hits one mALB1 (45 ACh, 3 of 8810). Type-LH hop-1 hits 93 of 93. DA1 onto type-LH that hit those 93: 161 cells, 8425. Those 161 onto the 93: 3301, covering 90 cells and 8773 of 8810. Hop: DA1_PN -> type-LH -> mAL_GABA_pre. On this tree, DA1-driven means any DA1 synapse (f_i > 0). @de95257, logs/drive_s1.json.

Fraction. Frozen 161 IDs from @de95257. Count in: DA1 onto those 161 is 8425. Count out: those 161 onto the 93 is 3301. Incoming weight by pre tag: other 155,526, LH 72,266, other_uPN 33,072, multi_PN 19,140, DA1_PN 8,425, mAL 4,234. f-bar = 0.099172 (all incoming). f-bar^PN = 0.702674 (uniglomerular PNs). LHAV4c2: 2 cells, 1012 of 3301, 5 DA1 synapses, f-bar 0.002537, f-bar^PN 1.0 (5/5). LH007m: 814 of 3301, 1714 DA1, f-bar^PN 0.728904. LH008m: 360 of 3301, 2851 DA1, f-bar^PN 0.73711. @01155c3, logs/lh_frac_s1.json.

3. Stubbed flags

--unfreeze, --female-brain-icarus, and --n 1000 stay stubbed on every tree in this chain.

4. Replay

Each lock is one extract plus, where a battery exists, the frozen-object assay.

Tree	Extract	Lock
fly_icarus	data/templates/male_p1.json	logs/p1_contact_s1.json, logs/p1_da1_dose_s1.json
fly_p1_sign	scripts/extract_p1_weights.py	logs/p1_sign_s1.json
fly_p1_mal	scripts/extract_p1_weights.py	data/templates/extract.json; do not restamp logs/p1_mal_s1.json
fly_mal_drive	scripts/extract_drive.py	logs/drive_s1.json
fly_lh_frac	scripts/extract_frac.py	logs/lh_frac_s1.json; IDs in data/frozen/ids.json

How to run: in each tree, .venv/bin/python -m pytest. Rebuilds, where allowed, are the extract scripts above. Do not overwrite the listed lock files.

5. Repo table

Tree	SHA	Allowed sentence	Lock file
fly_icarus	2cf5fd6	In this 11-cell slice, HD-on / cVA-off is necessary and sufficient for 3 matching 1. Morphology, pin, and feminized cuticle are not. Schema W_crit=-1.5262.	logs/p1_contact_s1.json
fly_p1_sign	45aa064	Hop-1 DA1 is +62 ACh. LC10a +606. ppk23 hop-1 is 0. hd_on_cva_off_ns is false.	logs/p1_sign_s1.json
fly_p1_mal	e16856c	Hop-1 mAL GABA onto these 88 is large enough to pass W_crit if assigned to cVA. DA1 hop-1 onto mAL_m* is 0; onto two mALB1 cells it is 96 ACh.	logs/p1_mal_s1.json
fly_p1_mal sentence	41437dc	Same allowed sentence as the row above.	README of fly_p1_mal
fly_mal_drive	de95257	Yes at hop-2. DA1 hop-1 is 3 of 8810. DA1-driven type-LH accounts for 8773 of 8810. Hop: DA1_PN -> type-LH -> mAL_GABA_pre.	logs/drive_s1.json

Tree	SHA	Allowed sentence	Lock file
fly_lh_frac	01155c3	f-bar=0.099172. f-bar^PN=0.702674. LHAV4c2 carries 1012 of 3301 with 5 DA1 synapses.	logs/lh_frac_s1.json

Figure 1 (figures/stack.png): object tags to the 11-cell row to hop-1 counts to hop-2 LH to the f table.

Sources (2026-09-20)

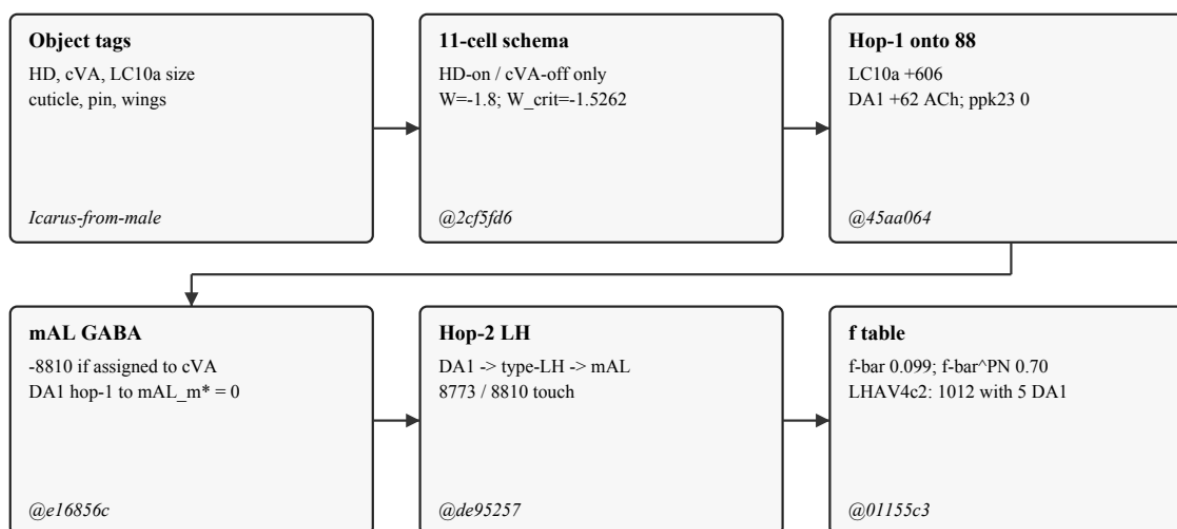
Records checked on Crossref, 2026-09-20. Female parent count is FlyWire. Male hop-1 counts are MaleCNS. The 11-cell row is a published-sign schema, not a hop-count extract.

Berg, S., Beckett, I. R., Costa, M., Schlegel, P., Januszewski, M., Marin, E. C., Nern, A., et al. (2026). Sexual dimorphism in the complete *Drosophila* male central nervous system connectome. *Cell*, 189(18), 5504-5526.e15. <https://doi.org/10.1016/j.cell.2026.08.015>

Dorkenwald, S., Matsliah, A., Sterling, A. R., Schlegel, P., Yu, S. C., McKellar, C. E., Lin, A., et al. (2024). Neuronal wiring diagram of an adult brain. *Nature*, 634(8032), 124-138. <https://doi.org/10.1038/s41586-024-07558-y>

Index (pointers only): <https://gist.github.com/martialsystems/12835f747d6360781f3cc7f91f243178>

Figure 1. Object tags to schema row to hop-1 counts to hop-2 LH to the f table (2026-09-16).



Locks stay on the trees. Pointer stack, not a new measurement.

Pointer stack across the locked trees. Not a new measurement.

Revisions (2026-09-16)

2026-09-16: first note from the locked trees. Same day: dropped the Not-stack; sentences taken from the lock files. Index gist 12835f74 left unedited.

2026-09-20: sources for MaleCNS (Berg et al., 2026) and FlyWire (Dorkenwald et al., 2024).